A BEVY 0.20 ARCADE TRIBUTE

ELEVATOR ACTION

RED DOOR PROTOCOL

Ride the elevators. Crack the red doors. Take the files. Get out before the night desk sends the whole floor after you.

A/D MOVE J SHOOT K JUMP W RIDE

3 STAGES 8 FLOORS 2 ELEVATORS 1 WAY OUT

github.com/orbitmechanic/elevator-action-bevy

Rust / Bevy 0.20.1 · Krita pixel art · original chiptune soundtrack



STAGE 1 - THE HOTEL MERIDIAN

AGENT 17

1983 - 2026

THE BRIEF

A COMPLETE ARCADE GAME

Rebuild the 1983 elevator-spy loop as a complete, playable 2D arcade game — not a tech demo.

Descend a cutaway hotel. Call and ride two elevators. Crack the red document doors. Dodge the night desk's agents. Collect every file, then reach the getaway car.

3	8	2
STAGES	FLOORS	ELEVATORS

One screen. Three escalating assignments.
No save points, no continues — just three lives.



STAGE 1 — THE HOTEL MERIDIAN

ORIGINAL ART / ORIGINAL AUDIO / NO ARCADE ROMS

THE LOOP

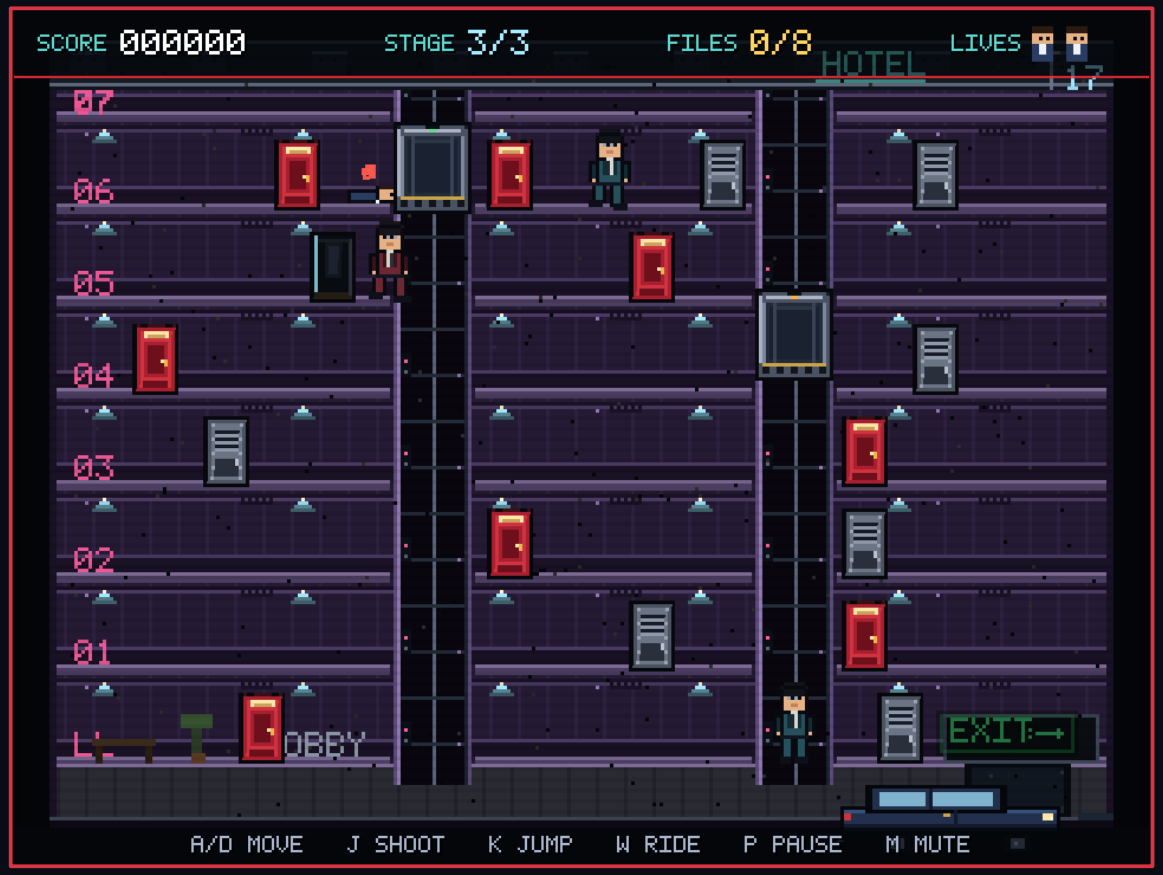
SIX FILES. NO EXIT.

01
CALL THE CAR
W at a shaft summons an elevator. W again boards it once it docks.

02
RIDE THE FLOORS
W/S picks the next stop. A/D steps off when the car is docked.

03
CRACK THE RED DOORS
Walk into a red door to take its file: +500. Collect them all to open the exit.

04
REACH THE CAR
Walk into the getaway car at the bottom right. Bonus scales with lives.



STAGE 3 — THE RED DOOR PROTOCOL

RED = FILES. GRAY = AGENTS. THE CAR OPENS ONLY WHEN EVERY FILE IS YOURS.

FEEL

DUCK. JUMP. SHOOT.

MOVE	A / D or ← / →
SHOOT	J / X (hold to autofire)
JUMP	K / Space
DUCK	S / ↓ (dodges chest shots)
ELEVATOR	W / ↑ call, board, next floor
PAUSE / MUTE	P or Esc / M

THE FEEL

One-hit deaths with 2.4 s of spawn protection keep mistakes legible. Ducking swaps to a 10 px hitbox at the feet, so chest-height bullets pass overhead. Agents telegraph for 0.55–0.9 s before firing.



MUZZLE FLASH. TWO ROUNDS. AN OPEN RED DOOR

UNDER THE HOOD

BEVY ECS, TOP TO BOTTOM

STATES
Title · InGame
GameOver · Victory

PLAYER
move · jump · duck
shoot · ride · respawn

ENEMY
floor-aware chase
call · board · shoot

ELEVATOR
CarState resource
position · target · riders

COMBAT
bullets · particles
hits · doors · exit

LEVEL
spawn · waves
transitions

UI
HUD · banners
pause · screens

WORLD
layout · levels
pure math

```
COMPONENTS   Player · Enemy · Door · Bullet · ElevatorCar
RESOURCES    RunState · Elevators · Fx · PendingLevel
MESSAGES     PlaySfx
SCHEDULES    Startup · Update · OnEnter(state)
```

ONE PLUGIN : 8 MODULES : NO UNSAFE : NO CUSTOM RENDERER

Systems are chained in a single Update schedule, gated by state and by a Normal-phase run condition. Commands, messages and despawn-on-exit markers keep level teardown and restarts clean.

SIMULATION

THE ELEVATOR IS THE LEVEL

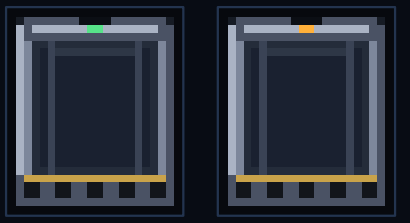
```rust
pub struct CarState {
    pub bottom: f32,
    pub target_floor: usize,
    pub moving: bool,
    pub riders: u32,
}
fn update(&mut self, dt: f32) -> Option<usize> {
    let target = car_bottom_for_floor(self.target_floor);
    let delta = target - self.bottom;
    let step = CAR_SPEED * dt;
    if delta.abs() <= step {
        self.bottom = target;     // dock
        self.moving = false;
        Some(self.target_floor)
    } else {
        self.bottom += delta.signum() * step;
        None
    }
}
```

SRC/WORLD.RS

WHAT MAKES IT WORK

- request() retargets a moving car, so a rider can change floors mid-trip.
- 44 px/s and a 26 px floor pitch: the ride reads as deliberate, not floaty.
- docked_floor() is the single truth for boarding, stepping off, and enemy AI.
- A riders counter stops agents from calling a car out from under the player.
- Agents walk to a shaft, call, board, retarget, and dismount — the same rules as the player.

TESTS CAR_RIDES_AND_DOCKS : CAR_DOES_NOT_MOVE_WHEN_ALREADY_THERE



IDLE MOVING

ART PIPELINE

PIXELS FROM SCRATCH



1× SOURCE ART — 7 PLAYER FRAMES, 2 AGENT SHEETS, DOORS, CARS, HUD

KRITA 6 / PYKRITA

Every sprite and all three backdrops are drawn at 1× into QImages, pushed through Krita layers, and exported as PNG. Editable .kra sources ship in assets/art/.

A 5×7 FONT, TYPESET IN CODE

A hand-built glyph table is expanded to a real TTF with fontTools, then loaded by Bevy for crisp HUD and menu text.

INKSCAPE

The README banner and social preview are vector SVG, rendered to a 1280×640 PNG.

NO EXTERNAL ART

No arcade rips, no stock sprites, no sample packs. Backgrounds are midnight, amber, and violet themes built from the same floor plan.



THREE BUILDINGS, ONE FLOOR PLAN

A 16-SECOND SPY LOOP



ORIGINAL CHIPTUNE — SQUARE, TRIANGLE AND NOISE, RENDERED TO WAV

- 11 cues plus the loop: pistol, enemy shot, hit, pickup, death, ding, door, call, clear, game over, victory.
- 22.05 kHz mono; all synthesized in pure Python with the standard library.
- One GlobalVolume resource handles mute without touching individual sinks.



CRISP HUD AT 3× — CUSTOM FONT, NO ANTIALIASING

320 × 240

VIRTUAL ARCADE SCREEN

rendered at 3× in a fixed 960×720 window

NEAREST

SAMPLING + INTEGER SNAP

no blurry pixels, ever



VERIFICATION

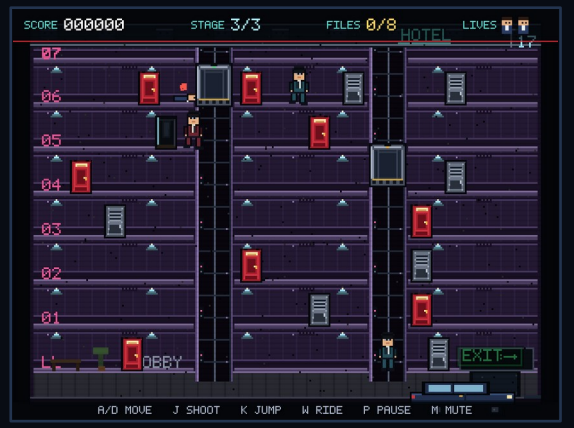
TRUST. BUT VERIFY



TITLE



STAGE 1



STAGE 3



PAUSE

EVERY IMAGE IN THIS DECK WAS CAPTURED FROM THE RUNNING BINARY.

FMT	LINT	TESTS	CAPTURE	DEMO
cargo fmt --check	clippy -D warnings	15 passing	self-screenshot	scripted input

Unit tests cover pure elevator/collision math and every level layout. Integration tests drive full ECS apps for document pickup, damage/life loss, respawn, stage advance, victory transition, and the complete level-spawn pipeline. Screenshot mode waits for the scene to settle, saves the framebuffer, and exits — the loop that caught UI overlaps, font fallback squares, and HUD/playfield collisions.

SHIP

PLAY IT

A complete, original arcade game in Rust.
Clone it, build it, crack some red doors.

```
git clone https://github.com/orbitmechanic/elevator-action-bevy
cd elevator-action-bevy
cargo run
cargo test
cargo clippy --all-targets -- -D warnings
```

MIT LICENSE : ORIGINAL ART + AUDIO : BEVY 0.20.1

Built with Krita, Inkscape, LibreOffice and a lot of pixel counting. Thanks for playing.

SCAN TO PLAY

GITHUB.COM/ORBITMECHANIC

PRESS ENTER TO START